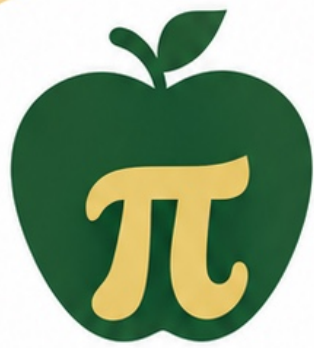




SCHOOL OF MATH

Think • Learn • Grow

Small Steps Big Futures

DECIMALS — TO THE TENTHS — AND HUNDREDTHS

A School of Math class on place value, reading decimals, comparing decimals, number lines, money, and measurement



Build confidence
with decimals



Connect math
to daily life



Develop precision
in money and
measurement



Clear explanations •
Strong foundations •
Engaging learning

Real Math. Brighter Futures.



schoolofmath.us

SAME
NUMBERS.
BRIGHTER
FUTURES.



**SCHOOL
OF MATH**

REAL MATH.
BRIGHT PEOPLE.
A BRIGHTER WORLD.

*Math Builds
What's Next*

PEOPLE
IDEAS
CULTURES
SOLVE
TOGETHER

LEARN
PRACTICE
APPLY
GROW

MATH
CONNECTS
OUR WORLD

TOPIC 1

Introduction to Decimals

Decimals help us describe parts of a whole using base-ten place value.

1 What is a Decimal?

A decimal is a number that uses a decimal point to show a whole number and parts of a whole.

Examples:

0.5

1.2

3.45

2 The Theory

- The digits to the left of the decimal point show whole numbers.
- The digits to the right show parts of a whole.
- Tenths means 1 out of 10 equal parts: **0.1**
- Hundredths means 1 out of 100 equal parts: **0.01**
- Decimals are based on the base-10 number system.

*Small
parts make
big things
possible!*

3 Visual Understanding

Place Value for a Decimal

Ones	Decimal Point	Tenths	Hundredths
4	.	3	7
↓		↓	↓
4 = 4 ones		3 = 3 tenths (3 out of 10)	7 = 7 hundredths (7 out of 100)

So, 4.37 means 4 ones, 3 tenths, and 7 hundredths.

Visual Models

Tenths (0.5)



0.5 means 5 out of 10 equal parts are shaded.

Hundredths (0.46)



In a race, 0.46 seconds is less than half a second. That's 46 hundredths of a second!

4 Real-Life Examples

Decimals show up everywhere!

Money



\$2.75

You might pay \$2.75 for a snack.

Measurement



1.5 meters

A desk might be 1.5 meters long.

Time in Sports



9.8 seconds

A sprinter might run 100 meters in 9.8 seconds.

Temperature



98.6°F

A normal human body temperature is 98.6°F.

5 Decimals Around the World



Decimals are used in science, money, sports, and measurement all over the world.

In many countries, people write a decimal with a comma instead of a point, so 3.5 may appear as 3,5. The metric system also uses decimals to describe length, mass, and volume.

*Different places.
Same ideas.
A brighter world
through math.*

MATH IN REAL LIFE.
OPPORTUNITIES IN REAL PEOPLE.

NUMBERS BRING US TOGETHER

BIGGER QUESTIONS.
BRIGHTER FUTURES.

Decimals to the Tenths and Hundredths

Image 1 practice: Introduction to decimals

Q1. Which number is written with a decimal point: 8, 24, 6.32, or 150?

Q2. In 0.8, which place does the 8 occupy, and what fraction does it represent?

Q3. In 4.06, which digit is in the hundredths place? What is its value?

Q4. Separate 8.24 into its whole-number part and fractional part. Write an addition sentence.

Q5. A student writes 6.4 for six whole units and four hundredths. Correct the number and explain the role of zero.



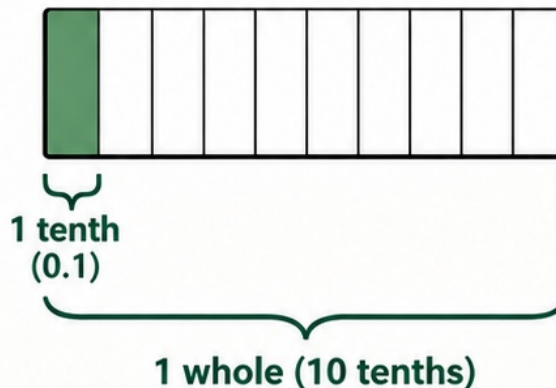
Small Steps
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Tenths: Dividing a Whole into 10 Equal Parts

A **tenth** is one of **10 equal parts** of a whole. When we divide something into 10 equal parts, each part is called a **tenth** and can be written as a fraction or a decimal.

$$\frac{1}{10} = 0.1$$

↑
One tenth



**10 tenths
make 1 whole.**

$$\frac{10}{10} = 1$$

$$1.0 = 1$$

$$0.1 + 0.1 + \dots + 0.1 \text{ (10 times)} = 1$$

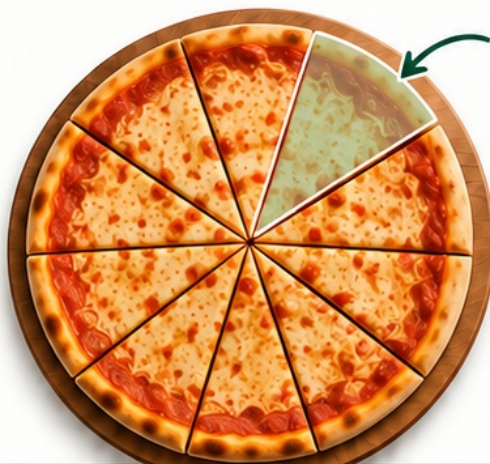
More Examples

$$\frac{2}{10} = 0.2$$

$$\frac{3}{10} = 0.3$$

$$\frac{7}{10} = 0.7$$

$$\frac{9}{10} = 0.9$$



One slice is
1 tenth (0.1)
of the pizza.

A Bit of History

The decimal system we use today developed from the work of mathematicians in India and was later brought to Europe. Using tenths, hundredths, and powers of 10 made calculations easier in science, trade, and everyday life — from measuring land to calculating money.



Simon Stevin
(1548–1620)
helped popularize
decimal notation
in Europe.

“Mathematics is the language in which God has written the universe.”
— Galileo Galilei



HIGHER
THAN GRADES

Decimals to the Tenths and Hundredths

Image 2 practice: Tenths: ten equal parts

Q6. Write $\frac{6}{10}$ as a decimal.

Q7. Write 0.8 as a fraction with denominator 10.

Q8. A strip has 10 equal sections. Five are colored. What decimal of the strip is uncolored?

Q9. A container holds 1.0 liter when full. It contains 0.6 liter. How many tenths of a liter are needed to fill it?

Q10. A path is divided into 10 equal sections. A child walks 2 sections, then 4 more. What decimal of the path remains, and what decimal was walked?



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Hundredths: Dividing a Whole into 100 Equal Parts

A **hundredth** is one of **100 equal parts** of a whole. When we divide something into 100 equal parts, each part is called a **hundredth** and can be written as a fraction or a decimal.

1. Core Idea

A **hundredth** is one of 100 equal parts of a whole.

$$\frac{1}{100} = 0.01$$

More examples:

$$\frac{25}{100} = 0.25 \quad \left| \quad \frac{37}{100} = 0.37 \quad \left| \quad \frac{8}{100} = 0.08 \right.$$

2. The Theory

The first digit to the right of the decimal point is the **tenths place** and the second digit is the **hundredths place**.



In 4.37, the 3 means **3 tenths** and the 7 means **7 hundredths**.

0.37 means **37 hundredths**.

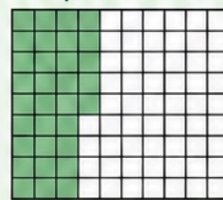
3. Visual Understanding

Place-Value Chart

Ones	Decimal Point	Tenths	Hundredths
4	.	3	7

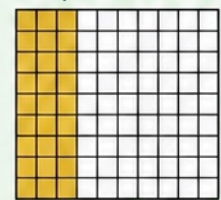
4.37 = 4 ones, 3 tenths, and 7 hundredths.

0.35 (35 hundredths)



37 out of 100 squares are shaded. This shows 0.35.

0.30 (30 hundredths)



25 out of 100 squares are shaded. This shows 0.30.

4. Real-Life Examples

Money



\$0.37 = 37 cents

Measurement



1.25 meters

Sports Timing



10.25 seconds

Science / Temperature



98.60°F

We can measure temperature to hundredths of a degree.

5. Around the World



Decimals are used around the world in money, science, and the metric system. Some countries write decimals with a comma instead of a point, so 3.25 may appear as 3,25.

6. A Bit of Math History

The decimal system developed over time through the work of mathematicians from different cultures, including India and later Europe. Decimals help people measure and calculate more precisely in science, trade, and everyday life.



Brahmagupta
(c. 598–668)

An Indian mathematician who helped develop our number system.

"Math helps us describe the world with precision."

— Unknown



THINK • LEARN • GROW

Decimals to the Tenths and Hundredths

Image 3 practice: Hundredths: one hundred equal parts

Q11. Write $46/100$ as a decimal.

Q12. Write 0.72 as a fraction with denominator 100.

Q13. Write $9/100$ as a decimal. Explain why 0.9 would be different.

Q14. A hundred-square grid has 52 shaded squares. Express the shaded amount in hundredths, as a decimal, and as tenths plus hundredths.

Q15. A hundred-square grid has 64 shaded squares. How many more squares must be shaded to show 0.80? What decimal of the grid is added?



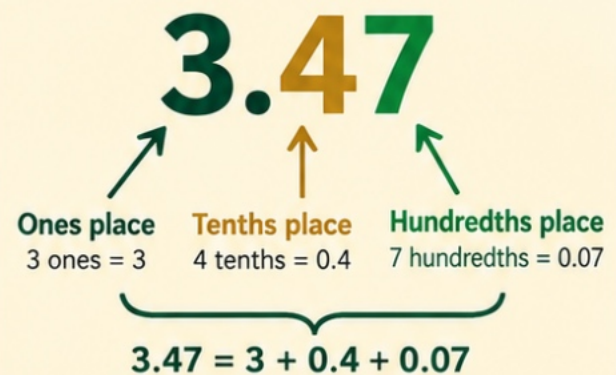
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Decimal Place Value

Each digit in a decimal number has a value based on its place.

In a decimal number, the digits to the right of the decimal point represent parts of a whole. The first digit to the right is the **tenths place**, and the second digit is the **hundredths place**.

Ones		Tenths	Hundredths
3	.	4	7
3 ones (whole numbers)		4 tenths ($4 \times 0.1 = 0.4$)	7 hundredths ($7 \times 0.01 = 0.07$)



More Examples

Look at the value of each digit.

5.23

5 ones
2 tenths
3 hundredths

$$5.23 = 5 + 0.2 + 0.03$$

0.81

0 ones
8 tenths
1 hundredth

$$0.81 = 0 + 0.8 + 0.01$$

12.06

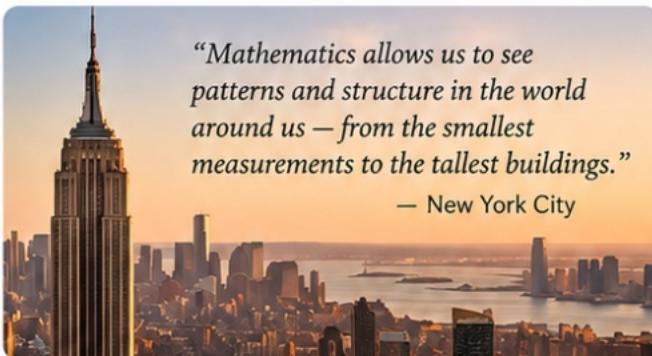
12 ones
0 tenths
6 hundredths

$$12.06 = 12 + 0 + 0.06$$

7.90

7 ones
9 tenths
0 hundredths

$$7.90 = 7 + 0.9 + 0$$



Real-World Connection

Decimals are everywhere! We see them in money, measurements, sports, and science.

- The Empire State Building is 1,454.33 feet tall.
- A pencil is 0.17 meters long.
- A gallon of milk costs \$3.49.
- A runner finished the race in 12.06 seconds.



"If I have seen further, it is by standing on the shoulders of giants."
 — Isaac Newton



HIGHER
THAN GRADES

Decimals to the Tenths and Hundredths

Image 4 practice: Decimal place value

Q16. What is the value of the 6 in 2.64?

Q17. Write 0.92 in expanded form.

Q18. Write 14.08 in expanded form, showing the tens, ones, and hundredths.

Q19. Are 6.5 and 6.50 equal? Explain using hundredths.

Q20. Use the digits 2, 5, and 8 once each to make the greatest number with one digit before the decimal point and two after it. Write its expanded form.



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Reading and Writing Decimals

Same number.
Different forms!

We can read and write decimals in different ways. A decimal can be written in **standard form** (using digits), in **words**, or as a **fraction** (when the denominator is 10 or 100).

Standard Form	Word Form	Fraction Form
0.3	three tenths	$\frac{3}{10}$
0.7	seven tenths	$\frac{7}{10}$
0.25	twenty-five hundredths	$\frac{25}{100}$
0.68	sixty-eight hundredths	$\frac{68}{100}$

How to Read Decimals

- 1 Read the whole number part (the number before the decimal point).
- 2 Read the decimal part as tenths or hundredths, depending on the last digit.
- 3 Use a hyphen between the whole number and the decimal part.

Examples

- 1.4 → one and four tenths
 2.17 → two and seventeen hundredths
 0.9 → nine tenths
 0.05 → five hundredths
 6.82 → six and eighty-two hundredths

Write These Decimals in Words

- a) **0.6** → _____
 b) **0.28** → _____
 c) **3.4** → _____
 d) **5.09** → _____

Write These in Standard Form

- a) nine tenths → _____
 b) forty-two hundredths → _____
 c) seven and three tenths → _____
 d) twelve and eight hundredths → _____



"Numbers are a universal language that help us describe the world around us."

— Unknown

Real-World Connection

We read and write decimals every day:

- Prices: \$2.75
- Measurements: 1.62 meters
- Sports: a race time of 9.58 seconds
- Science: 0.07 grams of a substance



SAME MATH
A BRIGHTER WORLD

Decimals to the Tenths and Hundredths

Image 5 practice: Reading and writing decimals

Q21. Write 0.4 in word form.

Q22. Write 0.56 in word form.

Q23. Write 'eight and two tenths' in standard decimal form.

Q24. Write 'six and four hundredths' in standard decimal form and expanded form.

Q25. Maya writes 9.7 for 'nine and seven hundredths.' Correct her number, write it as a whole number plus a fraction, and explain her mistake.



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Comparing and Ordering Decimals

We can compare decimals using place value. Look at the digits in the ones, tenths, and hundredths places, from left to right.

How to Compare Decimals

- 1 Compare the ones place. The greater ones place means a greater number.
- 2 If the ones place is the same, compare the tenths place.
- 3 If the tenths place is the same, compare the hundredths place.
- 4 You can also use a number line or think about which number is farther to the right.

Comparing Examples

$$0.6 > 0.4$$

6 tenths is greater than 4 tenths.

$$0.25 < 0.3$$

25 hundredths is less than 3 tenths (30 hundredths).

$$1.07 > 1.02$$

Same ones and tenths.
7 hundredths is greater than 2 hundredths.

$$0.58 = 0.58$$

The numbers are equal.

Ordering Decimals

To put decimals in order, compare them and arrange from least to greatest (smallest to largest) or from greatest to least (largest to smallest).

Order from least to greatest.

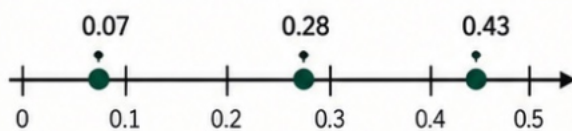
0.17 0.3 0.05 0.21
0.05 < 0.17 < 0.21 < 0.3

Order from greatest to least.

1.08 0.95 1.2 1.05
1.2 > 1.08 > 1.05 > 0.95

Using a Number Line

We can also compare decimals by looking at their positions on a number line.



Numbers to the right are greater.

Real-World Connection

We compare decimals every day!

- Prices: \$2.49 is less than \$2.75.
- Measurements: 1.3 m is greater than 0.95 m.
- Sports: A time of 12.08 seconds is less than 12.3 seconds.



"Comparison is the beginning of understanding."

— George Boole
(1815–1864)

Pioneer of modern algebra and logic



"Mathematics is not just about numbers, but about seeing the world more clearly."

— Unknown

Decimals to the Tenths and Hundredths

Image 6 practice: Comparing and ordering decimals

Q26. Fill the blank with $<$, $>$, or $=$: $0.8 \underline{\hspace{1cm}} 0.5$.

Q27. Fill the blank with $<$, $>$, or $=$: $0.46 \underline{\hspace{1cm}} 0.5$.

Q28. Compare 2.09 and 2.04. Explain which place decides the comparison.

Q29. Order these decimals from least to greatest: 0.62, 0.7, 0.09, 0.65.

Q30. Find every digit that makes $3.4\square$ greater than 3.46 but less than 3.50.



Small Steps
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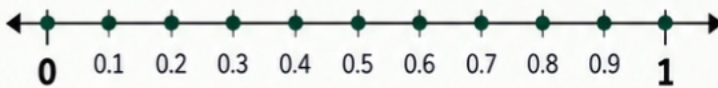
Decimals on a Number Line

Decimals help us find precise locations between whole numbers.

A number line shows numbers in order. We can use it to locate decimals such as **tenths** (0.1, 0.2, ...) and **hundredths** (0.01, 0.02, ...) between whole numbers.

Tenths on a Number Line

The distance from 0 to 1 is divided into 10 equal parts. Each point represents one tenth.



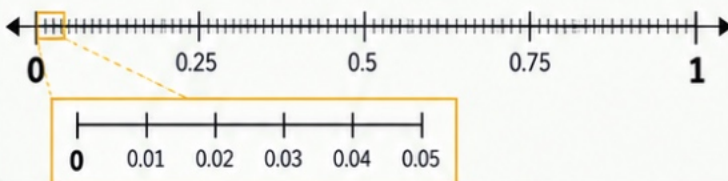
Examples

Locate each decimal on the number line.



Hundredths on a Number Line

The distance from 0 to 1 is divided into 100 equal parts. Each point represents one hundredth.



Examples

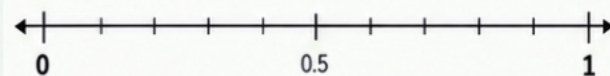
Locate each decimal on the number line.



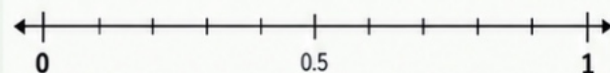
Your Turn

Estimate and mark the decimals on the number line.

a) 0.2 0.5 0.8



b) 0.06 0.33 0.71



Real-World Connection

We see decimals on number lines every day:

- Distances on a map (miles or kilometers)
- Measurements (meters, centimeters)
- Money (dollars and cents)
- Scores in sports (e.g., 9.85)
- Time (e.g., 0.75 seconds)



"The measure of a thing is what it is."
— Galileo Galilei



"Mathematics gives us a new perspective on the world."

— Maryam Mirzakhani
Fields Medalist in Mathematics

Decimals to the Tenths and Hundredths

Image 7 practice: Decimals on a number line

Q31. A number line runs from 0 to 1 in jumps of 0.1. How many jumps from 0 reach 0.5?

Q32. Which point lies farther right on a number line: 0.42 or 0.48?

Q33. Between which two consecutive tenths does 0.63 lie? Which tenth is closer?

Q34. Starting at 0.24, move three jumps of 0.1 to the right. Where do you land?

Q35. A point halfway between 0.52 and 0.60 is marked on a number line. What number is it? Explain in hundredths.



Small Steps
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Decimals in Real Life: Money and Measurement

We use decimals every day to work with money, measurements, and many real-life situations. Decimals help us be more precise and solve practical problems.

Money: Dollars and Cents

Money is written using decimals, with two digits representing cents (hundredths of a dollar).

\$2.75 means 2 dollars and 75 cents.



\$1.00



25¢
\$0.25



10¢
\$0.10



1¢
\$0.01

Measurement: Length

Decimals are used to show length, weight, and volume with greater precision.



This pencil is **3.47** centimeters long.
That means 3 centimeters and 47 hundredths of a centimeter.

Measurement: Weight and Volume



The strawberries weigh
0.85 kilograms
(85 hundredths of a kilogram).



The pitcher contains
0.50 liters
(50 hundredths of a liter).

Real-Life Word Problems

We can use decimals to solve everyday problems.

- 1 A notebook costs \$3.25. A pen costs \$1.40.
How much do they cost together?
 $\$3.25 + \$1.40 = \$4.65$
- 2 A piece of ribbon is 1.75 meters long.
You cut 0.60 meters. How much is left?
 $1.75 - 0.60 = 1.15 \text{ m}$
- 3 A water bottle holds 0.75 liters.
You drink 0.20 liters. How much is left?
 $0.75 - 0.20 = 0.55 \text{ L}$

Everyday Examples

- A gallon of milk costs \$3.49.
- A runner finished the race in 12.06 seconds.
- A smartphone is 0.78 cm thick.
- A recipe calls for 0.25 cups of sugar.
- A car travels 85.7 kilometers.

*"Mathematics is everywhere —
in the things we touch, the places we go,
and the world we build."*

— Unknown



REAL MATH. BRIGHTER FUTURES.

Decimals to the Tenths and Hundredths

Image 8 practice: Decimals in money and measurement

Q36. Write 'four dollars and sixty-two cents' as a decimal amount of dollars.

Q37. A ribbon measures 2.56 meters. How many whole meters and how many additional hundredths of a meter is that?

Q38. In a pretend shop, a notebook costs \$2.34 and an eraser costs \$1.25. What is the total cost?

Q39. A cord is 2.86 meters long. A student cuts off 1.24 meters. How much remains?

Q40. A jug contains 1.50 liters. Two cups each use 0.35 liter. How much liquid remains?

Decimals to the Tenths and Hundredths

Homework review: R1–R5

R1. Image 1: In 7.26, identify the whole-number part and the value of the digit 2.

R2. Image 2: A strip has 10 equal pieces and 8 are colored. Write the uncolored part as a fraction and a decimal.

R3. Image 3: A grid has 100 equal squares and 43 are shaded. Write the shaded amount as a decimal, then find how many squares are unshaded.

R4. Image 4: Write 3.82 in expanded form and explain why 3.820 has the same value.

R5. Image 5: Write 'eleven and six hundredths' as a decimal and as a whole number plus a fraction.

Decimals to the Tenths and Hundredths

Homework review: R6–R8

R6. Image 6: Order 0.84, 0.9, 0.79, and 0.87 from least to greatest.

R7. Image 7: Start at 0.32 on a number line and move five jumps of 0.01 to the right. Where do you land?

R8. Image 8: In a pretend shop, a folder costs \$2.46 and a pencil costs \$0.32. Find the total and the change from \$3.00.

Decimals to the Tenths and Hundredths

Homework word problems: W1–W5

W1. A garden row is divided into 10 equal sections. Lina plants flowers in 3 sections. What fraction and decimal of the row has flowers?

W2. A class mosaic has 100 equal tiles. Students color 58 tiles blue. What decimal of the mosaic is blue, and what decimal is not blue?

W3. A toy boat travels 2.07 meters. Write the distance in words and explain the value of the 7.

W4. Three seedlings measure 0.36 meter, 0.4 meter, and 0.29 meter. Order them from shortest to tallest and find the difference between the tallest and shortest.

W5. A robot starts at the 0.18-meter mark on a track. It moves forward two jumps of 0.1 meter and then four jumps of 0.01 meter. At what mark does it stop?

Decimals to the Tenths and Hundredths

Homework word problems: W6–W10

W6. In a pretend shop, a snack costs \$1.65 and a drink costs \$0.85. You pay with \$5.00. What is the total cost and how much change should you receive?

W7. A bottle contains 1.20 liters of water. A student pours out 0.35 liter, then adds 0.18 liter. How much water is in the bottle now?

W8. Two students complete the same course. Ava takes 14.26 seconds and Ben takes 14.6 seconds. Ava's time is 0.18 second faster than her previous attempt. Who wins, by how much, and what was Ava's previous time?

W9. A craft team has 2.50 meters of ribbon. It uses 0.65 meter for one project and 0.80 meter for another. It needs 1.20 meters for a third project. How much ribbon is left, and how much more must the team get?

W10. In a pretend shop, each craft kit needs a \$1.35 notebook and a \$0.65 pencil. A teacher buys supplies for three kits and pays \$10.00. Find the cost of one kit, the total cost, and the change.

Decimals to the Tenths and Hundredths

Homework word problems: W11–W12

W11. Four students need a pretend museum ticket and a snack each. Plan A charges \$2.45 per ticket and \$1.20 per snack. Plan B charges \$13.80 for all four tickets and snacks together. Which plan costs less, by how much, and what change is left from \$15.00 with that plan?

W12. A class joins three 1.25-meter ribbons end to end without overlap. It cuts four pieces of 0.65 meter and loses 0.05 meter in total to trimming. How much ribbon remains? Can it make two more 0.60-meter pieces, and if not, how much is it short?